

2008 级

一. 解:

初等因子: $\lambda, \lambda, (\lambda+2), (\lambda+2)^2$

行列式因子: $D_1(\lambda)=1$

$$D_2(\lambda)=\lambda(\lambda+2)$$

$$D_3(\lambda)=\lambda^2(\lambda+2)^2$$

故 $d_1(\lambda)=D_1(\lambda)=1$

$$d_2(\lambda)=\frac{D_2(\lambda)}{D_1(\lambda)}=\lambda(\lambda+2)$$

$$d_3(\lambda)=\frac{D_3(\lambda)}{D_2(\lambda)}=\lambda(\lambda+2)^2$$

$\therefore$ Smith 标准形为 $\begin{bmatrix} 1 & & \\ & \lambda(\lambda+2) & \\ & & \lambda(\lambda+2)^2 \end{bmatrix}$

二. 解

$$(1) |A-\lambda E| = \begin{vmatrix} 1-\lambda & 1 & 0 \\ -4 & 3-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = (\lambda-1)^2(2-\lambda)$$

对 $\lambda=1$,

$$A-E = \begin{bmatrix} -2 & 1 & 0 \\ -4 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore q = n - \text{rank}(A-E) = 1$$

故 Jordan 标准形为 $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

$$(2) \text{ 设 } p(x) = a_0 + a_1x + a_2x^2$$

$$\text{则 } p'(x) = a_1 + 2a_2x$$

$$p''(x) = 2a_2$$

$$\text{又 } p(2) = f(2) = a_0 + 2a_1 + 4a_2$$

$$p(1) = f(1) = a_0 + a_1 + a_2$$

$$p'(1) = f'(1) = a_1 + 2a_2$$

$$\therefore a_0 = f(2) - 2f'(1)$$

$$a_1 = 2f(1) - 2f(2) + 3f'(1)$$

$$a_2 = -f(1) + f(2) - f'(1)$$

$$\therefore f(A) = [f(2) - 2f'(1)]I + [2f(1) - 2f(2) + 3f'(1)]A + [-f(1) + f(2) - f'(1)]A^2$$

若 $f(A) = e^A$ 则 $f(x) = e^x, f(1) = e, f(2) = e^2, f'(1) = e$

$$\therefore e^A = \dots$$

若 $f(A) = \sin \frac{\pi}{2} A$ 则 $f(x) = \sin \frac{\pi}{2} x, f(1) = 1, f'(1) = 0, f(2) = 0$

$$\therefore \sin \frac{\pi}{2} A = 2A - A^2$$

三. 证明 见 P

$$(1) (AB)^H = B^H A^H = BA$$

四. 解

$$(1) A = \begin{bmatrix} -1 & 2 \\ 0 & 0 \\ 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{则 } B = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, C = [1 \ -2] \quad A = BC$$

$$(2) |A-\lambda E| = \begin{vmatrix} -1-\lambda & 2 \\ 0 & 0 \\ 1-\lambda & -2 \end{vmatrix} = -\lambda^2(\lambda-10)$$

$$AA^H = \begin{bmatrix} -1 & 2 \\ 0 & 0 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 5 & 0 & -5 \\ 0 & 0 & 0 \\ -5 & 0 & 5 \end{bmatrix}$$

$\lambda=10$ 时,

$$AA^H - 10E = \begin{bmatrix} -5 & 0 & -5 \\ 0 & -10 & 0 \\ -5 & 0 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ 单特征向量 $u_1 = (-1, 0, 1)^T, (-\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2})^T$

$\lambda=0$ 时,

$$AA^H \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{则 } u_2 = (\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2})^T, u_3 = (0, 1, 0)^T$$

$$\text{故 } U = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$\text{令 } V_1 = A^H U, \Delta^{-1} = (\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}})^T$$

$$\therefore V = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$\text{则 } A = U \begin{bmatrix} \sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} V^H$$

五. 证:

由 $AA^H = A^H A$ 存在酉矩阵 U , 使

$$U^H A U = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix}$$

$$|A - \lambda E| = \begin{vmatrix} -\lambda & 2 & 0 \\ 2 & -\lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix} = -\lambda(\lambda^2 + 4)$$

$$\therefore \lambda_1 = 0, \lambda_2 = 2i, \lambda_3 = -2i$$

$$\lambda = 0 \text{ 时, } A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \therefore \alpha_1 = (0, 0, 1)^T$$

$$\lambda = 2i \text{ 时, } A - 2iE = \begin{bmatrix} -2i & 2 & 0 \\ 2 & -2i & 0 \\ 0 & 0 & -2i \end{bmatrix} \sim \begin{bmatrix} -i & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \alpha_2 = (-\frac{i}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)^T$$

$$\lambda = -2i \text{ 时, } A + 2iE = \begin{bmatrix} 2i & 2 & 0 \\ 2 & 2i & 0 \\ 0 & 0 & 2i \end{bmatrix} \sim \begin{bmatrix} i & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \alpha_3 = (\frac{i}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)^T$$

$$G_1 = \alpha_1 \alpha_1^H = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$G_2 = \alpha_2 \alpha_2^H = \begin{bmatrix} \frac{1}{2} & -\frac{i}{2} & 0 \\ \frac{i}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_3 = \alpha_3 \alpha_3^H = \begin{bmatrix} \frac{1}{2} & \frac{i}{2} & 0 \\ -\frac{i}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{则 } A = 0 \cdot G_1 + 2i G_2 - 2i G_3$$

六. $P(A) = 0.5$

$$\begin{aligned} \sum_{k=1}^{\infty} k x^{k-1} &= \sum_{k=1}^{\infty} (x^k)' = (\sum_{k=1}^{\infty} x^k)' \\ &= (\frac{x}{1-x})', \quad |x| < 1 \\ &= \frac{1}{(1-x)^2}, \quad |x| < 1 \end{aligned}$$

$$R=1$$

(1) $P(A) < R$ 故 $\sum_{k=1}^{\infty} k A^{k-1}$ 收敛.

$$(2) \sum_{k=1}^{\infty} k A^{k-1} = (E - A)^{-2} = \begin{bmatrix} 4 & 16 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & \frac{25}{16} \end{bmatrix}$$

$$\text{七. 证: } \frac{dA(x)}{dx} = \begin{bmatrix} 0 & 2x \\ 2 & 0 \end{bmatrix}, \quad \frac{d^2 A(x)}{dx^2} = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

$$\int_0^t A(x) dx = \begin{bmatrix} t & \frac{2}{3} t^3 \\ t^2 & t \end{bmatrix}$$

$$\frac{d}{dt} \left[\int_0^t A(x) dx \right] = 3t^2 \cdot A(t^3) = \begin{bmatrix} 3t^2 & 3t^8 \\ 6t^5 & 3t^2 \end{bmatrix}$$

三. 证明:

A, B 都为 H 阵, $\therefore A^H = A, B^H = B$

又 A 正定, 则 $A = Q^H Q$

$$\begin{aligned} (1) |\lambda I - AB| &= |\lambda I - Q^H Q B (Q^H)^{-1} (Q^H)^{-1}| \\ &= |Q^H| |\lambda I - Q B Q^H| |Q^H|^{-1} \\ &= |\lambda I - Q B Q^H| \end{aligned}$$

$\therefore AB$ 的特征值与 $Q B Q^H$ 特征值相同

$$\text{又 } (Q B Q^H)^H = Q B^H Q^H = Q B Q^H$$

$\therefore Q B Q^H$ 为 H 阵, 其特征值为实数

$$\text{同理 } |\lambda I - BA| = |\lambda I - B Q^H Q|$$

$$= |\lambda I - Q^T Q B Q^H Q|$$

$$= |Q^T| |\lambda I - Q B Q^H| |Q|$$

$$= |\lambda I - Q B Q^H|$$

$\therefore BA$ 的特征值也为实数

$$(2) |\lambda A - B| = |A| |\lambda I - A^{-1} B|$$

$$= |A| |\lambda I - Q^T (Q^T)^H B Q^T Q|$$

$$= |A| |Q^T| |\lambda I - (Q^T)^H B Q^T| |Q|$$

$$= |A| |\lambda I - (Q^T)^H B Q^T|$$

$\therefore A$ 正定 $\therefore |A| > 0$

$\therefore |\lambda A - B| = 0$ 的根即为 $(Q^T)^H B Q^T$ 的特征值

又 $(Q^T)^H B Q^T$ 为 H 阵 $\Rightarrow$ 特征值为实数

$\therefore |\lambda A - B| = 0$ 的根为实数